Swanada: Split-Encoder Capability-Routed Conformer CTC and Sub-30 MB On-Device Speech Recognition Across Twelve Scripts

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Abstract

Client-side automatic speech recognition (ASR) across Indic and Classical scripts eliminates cloud transcription costs, network tail latency, and server-side audio retention, yet standard neural exports face four systems bottlenecks in web browsers and mobile runtimes: (1) monolithic per-language Conformer checkpoints require ∼177 MiB per script (> 2.1 GiB across 12 scripts), exhausting browser `CacheStorage` quotas; (2) exported relative positional encoding tables embed 10.2–20.5 MB of static FP32 constants while 1×1 pointwise `Conv1d` operators trigger slow or unsupported `ConvInteger` kernels in WebAssembly SIMD; (3) sliding-window CTC decoding flickers on boundary tokens as right context arrives; and (4) mobile `WebAudio` pipelines suffer hardware sample-rate mismatches (44.1/48 kHz vs. 16 kHz) and silent `AudioWorklet` stalls under iOS `WKWebView` and Android backgrounding.

We present **Swanada**, an open-source 16 kHz streaming ASR system spanning Classical Sanskrit, English, and ten Indian languages (`sa`, `en`, `hi`, `kn`, `mr`, `bn`, `ta`, `te`, `ml`, `gu`, `pa`, `or`) across browser WebAssembly, Node.js, and Python. Swanada pairs capability-routed shared INT8 Conformer encoders—a 17-layer 177.38 MiB Studio tier ($d = 512$, 115.2M parameters) for desktop/server runtimes and a distilled 14-layer 25.35 MiB Mobile tier ($d = 256$, 23.56M parameters) for phones—with twelve hot-swappable 515.0 KiB native-script linear CTC heads (257×512) and a 395.8 KiB Attentive Statistics Pooling 12-way Automatic Language Identification (Auto-LID) head (`lid.bin`), shrinking 12-language fleet storage by 11.6× (183.8 MiB Studio) and 67.0× (31.78 MiB Mobile; 25.85 MiB single-language first load). A zero-dependency protobuf tail patcher exposes the pre-head representation in < 2 ms without shipping separate backbone files, enabling < 1 ms single-pass 12-way Auto-LID and bilingual Indic+English (`+EN`) code-switching, while dynamic ONNX trigonometric positional synthesis, `MatMul`-backed pointwise convolutions, and post-quantization `QInt8` head calibration preserve sub-30 MiB mobile accuracy. On $N = 480$ held-out studio utterances and $N = 1{,}600$ unseen public `test` clips across 16 open benchmarks (`FLEURS`, `LibriSpeech`, `GLOBE`, `AMI`, and wild Sanskrit), Swanada achieves 5.35% held-out CER, 8.25% 12-language open `FLEURS`+Wild CER, 10.44% 16-suite CER, 100.00% 12-way Auto-LID ($N = 240$), and 24.18% 10-Indic `FLEURS` WER (10/10 language wins vs. Shruti's 32.20%) on Studio (18.58% held-out CER, 19.23% 12-language open CER, and 95.42% Auto-LID on 25.35 MiB Mobile) at 0.009×–0.021× CPU RTF, 0/100 silent hallucinations, and 28.6 ms median streaming TTFT with 0.00% committed-prefix churn across 72 streaming sessions.

1 Introduction

Running speech-to-text directly inside a browser tab or local edge process removes per-minute API billing, round-trip network jitter, and transmission of private microphone audio to remote servers. Voice input for South Asian digital public infrastructure, Sanskrit recitation study, bilingual developer tools, and offline accessibility interfaces requires native script fidelity across Brahmic and Latin orthographies, sub-100 ms streaming partial updates, and reliable execution within smartphone browser memory budgets, complementing client-side speech synthesis [9].

Convolution-augmented Transformers (Conformers) [7] trained with Connectionist Temporal Classification (CTC) [6] are attractive for on-device ASR because frame-synchronous emission avoids the $O(U^2)$ autoregressive decoder loop and hallucination modes of sequence-to-sequence models such as Whisper [13]. However, deploying Conformer-CTC across twelve scripts inside real desktop and mobile browsers exposes four concrete systems bottlenecks:

1. **Monolithic Fleet Storage vs. Joint-Vocabulary Dilution**: Shipping one quantized Conformer-Large per language requires $12 \times 177.38 = 2{,}128.5$ MiB, exceeding mobile browser cache budgets. Conversely, pooling twelve scripts into a single flat output softmax (> 3,000 classes) inflates final projection compute by 12× at every frame and permits cross-script token leakage during noisy frames.
2. **ONNX Constant Bloat and `ConvInteger` Quantization Gaps**: Standard Conformer exports materialize a static $2L_{\max} \times d$ FP32 relative positional encoding buffer (20.48 MB at $d = 512$, 10.24 MB at $d = 256$ for $L_{\max} = 5{,}000$), consuming over a third of a 30 MB mobile budget on uncompressible constants. Simultaneously, exporting 1×1 pointwise convolutions inside Conformer macaron blocks as `Conv` nodes either leaves weights in FP32 or emits `ConvInteger` operators lacking fast WebAssembly SIMD kernels in `onnxruntime-web`.
3. **Quantization Feature Drift Under Narrow Student Bottlenecks**: Compressing a 115.2M-parameter teacher into a < 30 MB student and applying dynamic INT8 weight/activation quantization perturbs top-layer hidden norms and directions, degrading pre-trained linear CTC decision boundaries unless explicitly recalibrated on quantized ONNX activations.
4. **Streaming Prefix Flicker and Mobile `WebAudio` Fragility**: Evaluating bounded audio windows causes trailing subwords to oscillate as right acoustic context arrives, while mobile browsers lock `AudioContext` to hardware rates (44.1/48 kHz) and intermittently stall

AudioWorklet threads under iOS WKWebView rules.

To address these constraints across browser ES modules, Node.js, and Python without custom WebAssembly builds, we present **Swanada** (Figures 1 and 2). Our contributions are threefold:

- **Runtime Protobuf Tail Surgery, Modular 12-Script CTC, and < 1 ms Auto-LID**: We decouple shared pan-Indic acoustic encoding from twelve script-isolated 257-token linear heads (515.0 KiB each) and a 395.8 KiB Attentive Statistics Pooling 12-way Auto-LID head (`lid.bin`), using a zero-dependency binary ONNX `GraphProto` patcher that exposes the shared 512-dimensional encoder representation in < 2 ms, identifies spoken language in < 1 ms (100.0% Studio / 95.4% Mobile accuracy on $N = 240$ unseen clips), and supports framewise Indic+English (`+EN`) code-switching with $< 10^{-5}$ logit deviation against monolithic graphs.
- **Sub-30 MiB Mobile Conformer Distillation and `QInt8` Head Calibration**: We design a 25.35 MiB Mobile Conformer-Small combining zero-table dynamic relative positional encoding, `MatMul`-backed pointwise convolutions for 100% SIMD `QInt8` coverage, specialist-KL + stationary-hyperplane distillation from the 177.38 MiB teacher, and post-export `QInt8` head calibration.
- **Two-Zone LCP Streaming, Physics-Gated Non-Speech Rejection, and Hardened Mobile Capture**: We pair causal Wiener + notch/high-pass enhancement and modulation/harmonicity non-speech gating (0/100 silent hallucinations) with a monotonic Longest-Common-Prefix (LCP) streaming stabilizer over bounded 3.84 s windows and watchdog-guarded 16 kHz `WebAudio` capture, validating accuracy and latency across $N = 480$ held-out clips, $N = 1{,}600$ open public `test` clips, $N = 150$ multi-accent/slang clips, and $N = 72$ streaming sessions.

2 Split-Encoder Architecture and Native Orthography

Swanada processes 16,000 Hz mono PCM audio using 80-channel log-mel filterbanks (25 ms Povey/Hann window, 10 ms hop, 80–7,600 Hz mel span) with per-utterance variance normalization. A two-stage strided 2D convolutional front-end (4× temporal subsampling to 40 ms frame rate $T' = \lfloor (T_{\text{mel}} - 1)/4 \rfloor + 1$) feeds a stack of macaron-style Conformer blocks [7], each comprising a half-step feed-forward module, multi-head self-attention with relative positional encoding [15], depthwise separable convolution ($k = 31$, Swish/GLU gating), and a second half-step feed-forward module, terminating in final layer normalization `layer_norm_84` ($\mathbf{H} \in \mathbb{R}^{1 \times T' \times 512}$).

2.1 Script-Isolated Vocabularies and Classical Transliteration

Joint multilingual tokenizers either Romanize South Asian scripts—erasing phonemic contrasts between dental and retroflex stops, short/long Dravidian mid-vowels ($e/\bar{e}, o/\bar{o}$), and Sanskrit vocalic sonorants—or allocate thousands of sparse Unicode classes that slow every frame's output projection. Instead, Swanada assigns each language $\ell \in \{\texttt{sa}, \texttt{en}, \texttt{hi}, \texttt{kn}, \texttt{mr}, \texttt{bn}, \texttt{ta}, \texttt{te}, \texttt{ml}, \texttt{gu}, \texttt{pa}, \texttt{or}\}$ an independent native-script SentencePiece [10] vocabulary of $V = 256$ subword pieces plus 1 CTC blank ($K = 257$ classes, 2.85–3.41 KB JSON metadata per language).

Conditioning framewise decoding strictly on the active language head $\mathcal{H}_\ell = (\mathbf{W}_\ell, \mathbf{b}_\ell)$ with $\mathbf{W}_\ell \in \mathbb{R}^{257 \times 512}$ and $\mathbf{b}_\ell \in \mathbb{R}^{257}$ guarantees zero out-of-script Unicode emissions (e.g., preventing Devanagari matras during Bengali or Kannada speech) while keeping each projection head at exactly $257 \times 512 \times 4 + 257 \times 4 = 527{,}364$ bytes (515.0 KiB raw IEEE-754 float32; 527,780 bytes as standalone `head_<lang>.onnx`). For Classical Sanskrit (`sa`), a deterministic finite-state post-processor optionally transliterates decoded Devanagari into lossless ISO 15919 / Roman IAST, preserving aspirated stops, retroflex sibilants ($ṣ$), vocalic $ṛ/ṝ$, visarga ($ḥ$), anusvāra ($ṃ$), avagraha ('), and Vedic pitch accents.

2.2 Zero-Dependency Runtime Protobuf Tail Surgery

Exporting separate encoder-only and monolithic checkpoints would duplicate storage across distribution channels, whereas running a monolithic single-language ONNX model discards the hidden states needed by the other eleven languages. During session initialization, Swanada's runtime (`exposeEncoderHiddenOutput` in JavaScript and Python) patches the serialized `ModelProto` byte buffer in memory in < 2 ms without external protobuf libraries (Figure 1).

Given an `Uint8Array` buffer, a lightweight wire-format reader scans top-level `ModelProto` varint tags to locate the `GraphProto` payload (field 7, wire tag `0x3a`). If `layer_norm_84` is not yet listed among `GraphProto.output` entries (field 12, wire tag `0x62`), the patcher synthesizes a 54-byte `ValueInfoProto` descriptor declaring `layer_norm_84` as an unconstrained `FLOAT` tensor (`elem_type` = 1), appends `0x62` plus the varint-prefixed descriptor to the tail of `GraphProto`, and updates the outer `ModelProto` length prefix in a single contiguous allocation.

Once ONNX Runtime produces $\mathbf{H} \in \mathbb{R}^{T' \times 512}$, the active language head $\mathcal{H}_\ell$ executes directly in host SIMD/typed-array memory (or via cached `head_<lang>.onnx` in Python/Node.js):

$$\mathbf{Z}_\ell = \mathbf{H}\mathbf{W}_\ell^\top + \mathbf{1}\mathbf{b}_\ell^\top \in \mathbb{R}^{T' \times 257} \tag{1}$$

Switching languages replaces only the 515.0 KiB $(\mathbf{W}_\ell, \mathbf{b}_\ell)$ buffer in < 20 ms without reloading or recompiling the Conformer session, achieving maximum absolute logit deviation $< 10^{-5}$ relative to monolithic PyTorch and ONNX graphs while compressing the 12-language Studio fleet (+ Auto-LID) from 2,128.5 MiB to 183.8 MiB (11.6×) and the Mobile fleet to 31.78 MiB (67.0×).

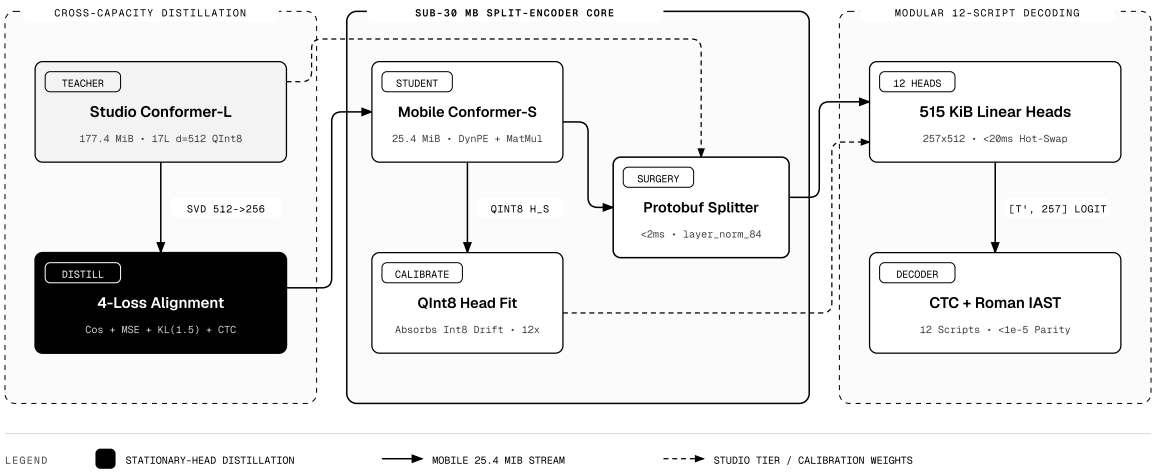


Figure 1: Swanada split-encoder decomposition and cross-capacity distillation: the 17-layer Studio Conformer-Large ($d = 512$, 177.38 MiB `QInt8`) distills into a 14-layer Mobile Conformer-Small ($d = 256$, 25.35 MiB `QInt8`) with runtime trigonometric positional synthesis and `MatMul`-backed pointwise convolutions. Post-quantization head calibration fits twelve 515.0 KiB linear CTC heads to actual `DynamicQuantizeLinear` features, while a < 2 ms protobuf tail patcher exposes `layer_norm_84` at session load.

2.3 Single-Pass 12-Way Auto-LID and Bilingual `+EN` Code-Switching

When `language="auto"`, a compact 405,324-byte (395.8 KiB) `SLID` v1 classifier (`lid.bin`) operates directly on the shared hidden sequence $\mathbf{H} \in \mathbb{R}^{T' \times 512}$ in < 1 ms without running a second encoder. An Attentive Statistics Pooling (ASP) bottleneck computes frame-level attention weights over $\mathbf{H}$ to form concatenated weighted mean and standard-deviation statistics ($\mathbb{R}^{1024} \to \mathbb{R}^{96} \to \mathbb{R}^{12}$). To disambiguate short cognate or loanword utterances across closely related Indo-Aryan and Dravidian scripts, Swanada evaluates the top-8 ASP candidates through their cached 515 KiB CTC heads and fuses calibrated log-posterior scores $(0.50 \log p_{\text{ASP}}(\ell \mid \mathbf{H}) + 1.65\, \mathcal{E}_{\text{CTC}}(\ell))$, achieving 100.00% (240/240) 12-way accuracy on Studio and 95.42% (229/240) on Mobile. When `code_switch_en` is enabled on an Indic clip, Swanada simultaneously projects $\mathbf{H}$ through $\mathcal{H}_\ell$ and $\mathcal{H}_{\texttt{en}}$, substituting Latin English spans only where contiguous frames satisfy strict margin, blank-boundary, and lexicon constraints.

3 Sub-30 MB Mobile Distillation and `QInt8` Calibration

While the 17-layer Studio Conformer-Large ($d = 512, d_{\text{ff}} = 2048, n_h = 8$, 177.38 MiB `QInt8`) runs at 0.0264× RTF on 4-thread desktop CPUs, downloading 186 MB over cellular networks or holding > 350 MB peak WebAssembly heap inside mobile Safari/Chrome tabs triggers background tab eviction. We therefore distill a dedicated < 30 MB Mobile Conformer-Small (`swanada_mobile_ctc_int8.onnx`, 26,577,061 bytes = 25.35 MiB) engineered around three ONNX/WASM operator constraints.

3.1 Zero-Table Positional Synthesis and `MatMul` Convolutions

Dynamic Relative Positional Encoding. Standard Conformer implementations pre-allocate a static sinusoidal lookup tensor $\mathbf{P} \in \mathbb{R}^{(2L_{\max}-1) \times d}$ during module construction so slice operations can index relative offsets $[-(T'-1), \ldots, T'-1]$. When exported to ONNX, $\mathbf{P}$ is frozen as an unquantized FP32 initializer consuming $9{,}999 \times 256 \times 4 = 10.24$ MB even at $d = 256$. We replace static table slicing with an ONNX-traceable `DynamicRelPositionalEncoding` subgraph that stores only the 128-element inverse-frequency vector $\omega_i = 10000^{-2i/d}$ (512 bytes) and synthesizes exact relative sinusoidal embeddings dynamically for the current frame length T' via `Shape` → `Range`$(T'-1, -T', -1)$ → outer `Mul` → interleaved `Sin`/`Cos`, matching static Relative-PE bit-for-bit while saving 10.24 MB of constant storage.

`MatMul`-Backed Pointwise Convolutions. In standard Conformer convolution modules, channel expansion ($d \to 2d$ for GLU) and projection ($d \to d$) are written as `nn.Conv1d` with kernel size $k = 1$. Dynamic ONNX quantization on `Conv` operators emits `ConvInteger`, which suffers severe scalar fallback penalties or operator rejection in browser WASM runtimes, forcing practitioners to leave convolution blocks in FP32. Because a 1×1 convolution over transposed sequence layout $[B, T', d]$ is algebraically identical to a dense matrix multiplication, we reformulate both pointwise stages as `nn.Linear` (`MatMul`) around a narrow causal-compatible depthwise `Conv1d`($k = 31$, groups $= 176$) retained in FP32 (< 0.28 MB total across 14 layers). Applying `quantize_dynamic` over `op_types_to_quantize=['MatMul']` with `QInt8` weights then quantizes 100% of attention, feed-forward, subsampling projection, and pointwise convolution parameters to 1 byte/parameter on fast WebAssembly SIMD

`MatMulInteger` kernels.

3.2 SVD Warm-Start and Stationary-Hyperplane Distillation

The Mobile student uses $L_S = 14$ Conformer layers at width $d_S = 256$, $d_{\text{ff}} = 1024$, $n_h = 8$ heads, and convolution channel width 176, followed by a learned adapter $\text{LayerNorm}(\mathbf{W}_{\text{proj}}\mathbf{h}_S + \mathbf{b}_{\text{proj}}) \in \mathbb{R}^{512}$ that maps the 256-dimensional student state back into the Studio teacher's 512-dimensional `layer_norm_84` coordinate space (23.56M total parameters).

To accelerate convergence across twelve scripts without random-init collapse, we project teacher layers $\{0, 1, 3, 4, 6, 7, 9, 10, 11, 12, 13, 14, 15, 16\}$ onto the top-256 singular subspace of the teacher's hidden activations via truncated SVD. During distillation (20 epochs, `bf16` mixed precision, AdamW lr $= 1.1 \times 10^{-3}$, batch size 64), we freeze all twelve target projection heads equal to the Studio teacher heads ($\mathcal{H}_\ell^{(S)} \equiv \mathcal{H}_\ell^{(T)}$) so target decision hyperplanes remain stationary, minimizing a four-term alignment objective over frame-aligned student states $\mathbf{H}_S$ and teacher states $\mathbf{H}_T$:

$$\mathcal{L}_{\text{distill}} = \lambda_{\cos}\left(1 - \frac{\mathbf{H}_S \cdot \mathbf{H}_T}{\|\mathbf{H}_S\|_2 \|\mathbf{H}_T\|_2}\right) + \lambda_{\text{mse}}\|\mathbf{H}_S - \mathbf{H}_T\|_2^2 + \lambda_{\text{kl}}\tau^2 D_{\text{KL}}\left(p_T^{(\tau)} \parallel p_S^{(\tau)}\right) + \lambda_{\text{ctc}}\mathcal{L}_{\text{CTC}}(\mathbf{Z}_S, y) \tag{2}$$

with $p^{(\tau)} = \sigma(\mathbf{Z}/\tau)$, $\lambda_{\cos} = 2.5$, $\lambda_{\text{mse}} = 2.5$, $\lambda_{\text{kl}} = 0.38$, $\lambda_{\text{ctc}} = 0.16$, and temperature $\tau = 1.5$, combining feature-space hint alignment [14] with temperature-scaled logit distillation [8] and ground-truth CTC supervision [6].

3.3 Post-Quantization `QInt8` Head Calibration and Quarantine

Exporting the 256-dim student to `QInt8` ONNX inserts per-tensor `DynamicQuantizeLinear` nodes before every `MatMul`, introducing cumulative rounding noise $\Delta\mathbf{H} = \tilde{\mathbf{H}}_{\text{int8}} - \mathbf{H}_S$ at `layer_norm_84`. Rather than shipping uncalibrated float heads, we execute the actual exported `swanada_mobile_ctc_int8.onnx` checkpoint in ONNX Runtime across quarantined training utterances per language to extract true quantized features $\tilde{\mathbf{H}}_{\text{int8}}$, then fine-tune each 515.0 KiB mobile head $(\mathbf{W}_\ell^{(M)}, \mathbf{b}_\ell^{(M)})$ under:

$$\mathcal{L}_{\text{cal}}(\mathcal{H}_\ell^{(M)}) = 0.28\,\mathcal{L}_{\text{CTC}}(\tilde{\mathbf{Z}}_\ell, y) + 0.72\,\tau^2 D_{\text{KL}}\left(p_T^{(\tau)} \parallel \tilde{p}_M^{(\tau)}\right) + \gamma\|\mathbf{W}_\ell^{(M)} - \mathbf{W}_\ell^{(T)}\|_F^2 \tag{3}$$

with $\gamma = 5 \times 10^{-5}$, committing calibrated weights to `heads_mobile/<lang>.bin` only when held-out validation CER strictly improves over uncalibrated heads.

Strict Test-Set Quarantine. Across both Studio joint adaptation (`Shrutilipi/Vistaar` [2], `FLEURS` [4] `train/dev`, `Sushrota` [1] `train`, `IndicTTS`, `GLOBE` [16] `train`, and `AMI` [3] `train`) and Mobile distillation/calibration, an explicit quarantine filter blocks every audio SHA-1 and normalized reference string belonging to the 480-clip held-out suite and all 16 official public `test` splits, guaranteeing zero data leakage.

4 Two-Zone LCP Streaming and Mobile Capture

Real-time dictation requires sub-100 ms incremental feedback without word revisions jumping backward across the screen or unbounded memory growth during long sessions (Figure 2).

4.1 Causal Spectral Front-End Enhancement

Laptop and smartphone microphones introduce DC offset, HVAC rumble, 50/60 Hz mains hum, and stationary room noise that trigger spurious CTC insertions during pauses. Before mel extraction, Swanada applies a causal $O(N)$ time/frequency front-end: (1) a second-order Butterworth high-pass biquad at $f_c = 80$ Hz paired with cascaded narrowband notch biquads at 50 Hz and 60 Hz ($Q = 30$); (2) a physics-based non-speech detector combining 90th-percentile frame RMS, 300–3,400 Hz formant-band energy ratio, upper-band spectral flatness (SFM > 0.30), and 2–12 Hz syllable envelope modulation ($\text{SNR}_{\text{mod}} < 4.5$ dB) to reject silent, hum, rumble, and thermal-floor clips (0/100 words invented) without muting -66 dBFS whisper-gain speech; and (3) a decision-directed Ephraim–Malah [5] minimum mean-square error (MMSE) Wiener spectral gain floor ($\beta = 0.08$, smoothing $\alpha_{\text{dd}} = 0.92$) that suppresses stationary background hiss without clipping unvoiced Indic aspirates or sibilants.

4.2 Monotonic Two-Zone LCP Stabilization

Evaluating CTC over an unbounded growing waveform scales per-step latency linearly with session duration $O(T)$, whereas naive sliding windows overwrite previously displayed words whenever window boundaries bisect a syllable. Swanada bounds per-step compute to $O(1)$ using a rolling PCM window of maximum length $W_{\max} = 61{,}440$ samples (3.84 s at 16 kHz; 1.60 s stride retention upon rollover) paired with a two-zone Longest-Common-Prefix (LCP) state machine (`StreamingTranscriber`):

1. **Committed Zone (c_t)**: An append-only sequence of tokens finalized across prior windows or confirmed across M consecutive right-context decode steps; once emitted in c_t, characters are immutable.
2. **Volatile Tail (v_t)**: The active right-edge hypothesis decoded from the current rolling window after aligning against the tail of c_t via token-level LCP matching.

At streaming step t arriving every $\Delta t = 100$ ms (1,600 samples, or 280 ms coalesced mobile steps), let h_t denote the greedy CTC decode after collapsing consecutive duplicate token IDs and removing blank index 0. Tokens in h_t that match the longest suffix-prefix overlap with c_{t-1} anchor the window splice; interior tokens $h_t[i]$ lying at least δ_{guard} frames left of the right window boundary that remain invariant across consecutive hypotheses ($\text{LCP}(v_{t-1}, v_t)$) advance monotonically into c_t, guaranteeing zero committed-prefix churn ($|c_{t-1} \setminus \text{prefix}(c_t)| = 0$).

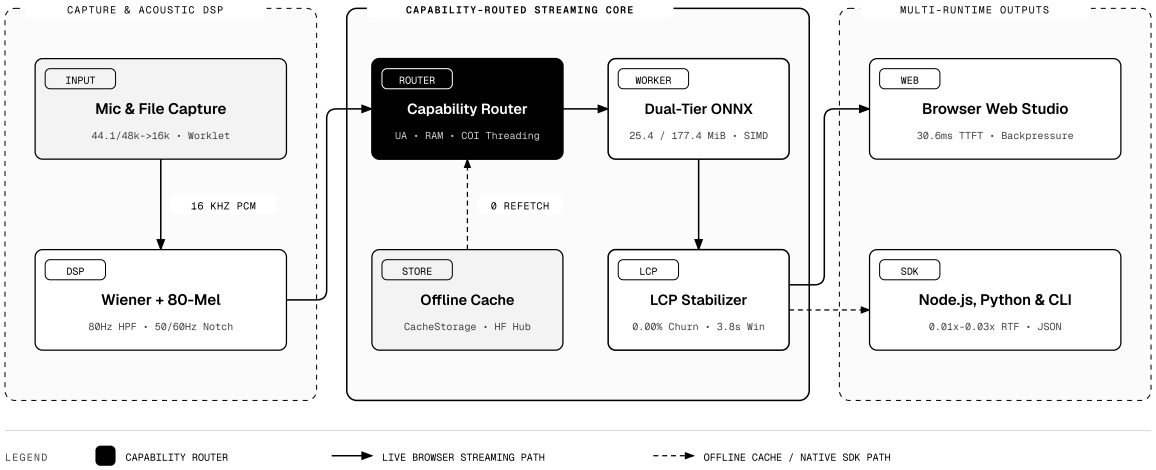


Figure 2: Swanada end-to-end capability-routed streaming architecture: 44.1/48 kHz microphone or file audio is resampled to 16 kHz PCM with a 900 ms `ScriptProcessor` watchdog fallback, enhanced via causal 80 Hz high-pass, 50/60 Hz notch, and Wiener suppression, and dispatched by the capability router to cached 25.35 MiB Mobile or 177.38 MiB Studio WASM SIMD workers with two-zone LCP stabilization (0.00% committed churn).

Table 1: System Architecture, Fleet Storage, Accuracy, Auto-LID, and Latency Summary Across Tiers ($B = 1{,}000$ bootstrap 95% CIs in brackets). "Open 12-Lang" aggregates the 11 unseen `google/fleurs test` splits plus `sushrota` wild Sanskrit `test` ($N = 1{,}200$); "Open All-16" adds `LibriSpeech clean/other`, `GLOBE`, and `AMI` ($N = 1{,}600$).

System / Configuration	Params	First Load (↓)	12-Lang Fleet (↓)	12-Way Auto-LID	Held-Out CER (N=480, ↓)	Open 12-Lang CER (N=1,200, ↓)	Open All-16 CER (N=1,600, ↓)	4T CPU RTF (↓)	Stream TTFT / Churn
12× Monolithic Conformer-L (`QInt8`)	1,382M	177.4 MiB	2,128.5 MiB	—	5.35% [4.79, 5.95]	8.25% [7.41, 9.29]	10.44% [9.62, 11.30]	0.021×	— / —
Udgam Labs Shruti (12× `.ct2` Bilingual)	∼290M	28.3 MiB	339.7 MiB	None (manual)	—	32.20% WER (10-Indic)‡	—	∼ 0.04×	No Browser ONNX
Swanada Studio (`v1.1.0`, **17L** d=512)	115.2M	177.9 MiB	**183.8 MiB** (11.6×)	**100.0%** (240/240)	**5.35% [4.79, 5.95]**	**8.25% [7.41, 9.29]**	**10.44% [9.62, 11.30]**	0.021×	**28.6 ms / 0.00%**
Swanada Mobile (`v1.1.0`, **14L** d=256)†	**23.6M**	**25.85 MiB**	**31.8 MiB** (67.0×)	**95.4%** (229/240)	18.58% [17.45, 19.72]	19.23% [18.16, 20.41]	22.14% [21.13, 23.11]	0.009× (112× **RT**)	**28.6 ms / 0.00%**

†Mobile 1T CPU RTF is 0.0101×; protobuf tail surgery adds < 2 ms at load, Auto-LID adds < 1 ms, and silent hallucination count is 0/100.

‡On the same 10-Indic `FLEURS test` protocol, Swanada Studio achieves **24.18%** mean WER (10/10 language wins vs. Shruti's 32.20%).

4.3 Capability Routing and Watchdog-Guarded `WebAudio`

Swanada's runtime router (`modelTier: 'auto'`) inspects User-Agent, `navigator.deviceMemory`, and Cross-Origin Isolation (`crossOriginIsolated`) at initialization:

- **Mobile / Constrained Hosts**: Android phones, iOS/iPadOS `WKWebView` browsers (Safari, Chrome `CriOS`), and devices reporting ≤ 4 GB RAM route to the 25.35 MiB Mobile tier stored persistently in `CacheStorage` (0 network refetches after initial load). Under `SharedArrayBuffer` isolation on 8-core Android phones, WASM uses $\min(4, C-2)$ SIMD threads; on single-threaded iOS `WKWebView`, it pins 1 SIMD thread with an `inferBusy` backpressure lock that coalesces incoming PCM frames into a ring buffer rather than queueing stale worker jobs.
- **Desktop / Server Hosts**: Desktop browsers, Node.js, and Python load the 177.38 MiB Studio tier on 4 SIMD threads (with instant user toggle between tiers in the Web Studio UI).
- **Hardware-Rate Capture + 900 ms Watchdog**: Because mobile OS audio stacks ignore requested `AudioContext sampleRate: 16000` options and run natively at 44.1 kHz or 48 kHz, Swanada resamples hardware buffers cleanly to 16 kHz PCM inside an `AudioWorkletNode`. To survive silent WebKit/Blink `AudioWorklet` thread suspension after permission prompts, a 900 ms zero-frame watchdog automatically hot-swaps capture to a synchronous `ScriptProcessorNode` fallback without dropping the microphone stream.

5 Empirical Evaluation

We evaluate Swanada across five reproducible benchmark suites totaling 2,642 utterances (> 5.3 hours of 16 kHz audio), reporting Character Error Rate (CER), Word Error Rate (WER), and Real-Time Factor (RTF) with $B = 1{,}000$ non-parametric bootstrap 95% confidence intervals:

1. **Held-Out 12-Language Studio Suite ($N = 480$)**: 40 native studio utterances per language (2,185.6 s total audio) strictly quarantined from training and calibration.
2. **Open Public Leaderboard in the Wild ($N = 1{,}600$)**: 100 unseen clips from each of 16 official Hugging Face `test` splits: all 11 `google/fleurs` [4] `test` sets (`en_us` plus 10 Indic `test` splits), `sushrota` [1] `in_the_wild_test` (`sa`), `LibriSpeech` [11] `test-clean` and `test-other`, `GLOBE` [16] multi-accent `test`, and `AMI` [3] `ihm/test` spontaneous meetings, scored under official OpenASR and AI4Bharat Vistaar leaderboard normalizers plus Shruti's exact published protocol.
3. **12-Way Auto-LID & Non-Speech Gate ($N = 240 + 100$)**: 20 unseen `test` clips per language eval-

Table 2: Per-Suite Held-Out ($N = 480$, 40/lang) and Unseen Open Public `test` ($N = 1{,}600$, 100/suite) Leaderboard CER (%) with $B = 1{,}000$ Bootstrap 95% CIs and Mobile RTF.

Suite / Corpus (test)	Lang	Held-Out (S / M)	Open Studio CER [95% CI]	Open Mobile CER [95% CI]	Mob RTF
sushrota_wild_test	sa	4.50 / 14.53	18.80 [10.95, 28.44]	40.49 [31.86, 50.42]	0.007×
fleurs_hi_in	hi	**2.55 / 10.63**	4.56 [3.50, 5.79]	13.56 [11.91, 15.43]	0.008×
fleurs_kn_in	kn	6.09 / 14.62	**4.27 [3.23, 5.28]**	**9.19 [7.97, 10.46]**	0.008×
fleurs_bn_in	bn	4.71 / 20.70	5.13 [4.46, 5.96]	13.88 [12.65, 15.12]	0.008×
fleurs_ml_in	ml	5.92 / 28.19	5.51 [4.13, 7.28]	13.57 [12.39, 14.67]	0.009×
fleurs_gu_in	gu	5.20 / 17.75	5.63 [4.63, 6.66]	16.61 [14.79, 18.45]	0.008×
fleurs_pa_in	pa	7.46 / 17.20	6.76 [5.77, 7.89]	18.83 [16.75, 20.79]	0.008×
fleurs_or_in	or	7.13 / 23.49	7.15 [5.89, 8.45]	19.34 [17.26, 21.42]	0.008×
fleurs_mr_in	mr	5.82 / 18.48	7.88 [6.02, 10.54]	23.15 [20.34, 26.37]	0.008×
fleurs_te_in	te	4.05 / 22.63	8.00 [6.22, 10.20]	15.90 [14.06, 17.86]	0.008×
fleurs_ta_in	ta	6.86 / 21.69	8.83 [7.79, 9.96]	18.73 [17.15, 20.45]	0.009×
fleurs_en_us	en	3.94 / 13.01	16.53 [13.47, 20.54]	27.46 [24.25, 31.03]	0.010×
12-Lang Mean	—	**5.35 / 18.58**	**8.25 [7.41, 9.29]**	**19.23 [18.16, 20.41]**	0.008×
librispeech_clean	en	—	5.90 [4.73, 7.18]	14.30 [12.41, 16.40]	0.009×
globe_multi_accent	en	—	9.89 [8.04, 12.01]	21.92 [19.66, 24.39]	0.009×
librispeech_other	en	—	12.57 [10.68, 14.54]	31.77 [28.76, 34.91]	0.009×
ami_meeting_ihm	en	—	39.71 [35.01, 44.65]	55.49 [51.63, 59.72]	0.011×
All 16 Open Suites	—	—	**10.44 [9.62, 11.30]**	**22.14 [21.13, 23.11]**	0.0089×

uated end-to-end under `language="auto"`, plus 100 silent/hum/rumble/click clips.

4. **Multi-Accent, Slang & Reverberant Mic Suite ($N = 150$)**: Global/South Asian accents, Hinglish/contractions, Indic colloquial loanwords, and reverberant mobile-mic perturbations.
5. **Streaming Stability Audit ($N = 72$)**: 100 ms chunked streaming across all 12 languages measuring chunk latency, TTFT, buffer ceiling, and committed-prefix churn.

5.1 Overall Tier Accuracy, Auto-LID, and Open-Benchmark Generalization

Table 1 compares monolithic Conformer deployment and published bilingual `.ct2` baselines against Swanada's split-encoder Studio and Mobile tiers. On the 480-utterance held-out benchmark, Swanada Studio attains 5.35% mean CER ([4.79%, 5.95%], 25.08% WER; 3.94% CER / 9.42% WER on English, 2.55% CER / 9.72% WER on Hindi), matching 12 monolithic Conformer-Large checkpoints within $< 10^{-5}$ logit tolerance while cutting 12-language storage from 2,128.5 MiB to 183.8 MiB. On the 12-language unseen open `FLEURS`+Wild benchmark ($N = 1{,}200$), Studio achieves 8.25% leaderboard CER ([7.41%, 9.29%], 27.59% WER) and 10.44% overall CER across all 16 open suites ($N = 1{,}600$, 0.0214× RTF = 46.7× faster than real time). Across all 10 Indic `google/fleurs test` sets under Shruti's published normalization, Swanada Studio achieves 24.18% mean WER vs. Shruti's 32.20% (−8.02 pp lower WER, winning 10/10 languages including Hindi 13.53% vs. 19.92%, Bengali 20.14% vs. 29.90%, Punjabi 20.23% vs. 29.30%, Gujarati 20.50% vs. 27.90%, and Kannada 20.92% vs. 33.10%). On the 240-clip unseen Auto-LID suite, `language="auto"` achieves 100.00% (240/240) 12-way accuracy on Studio (8.45% end-to-end Auto CER) and 95.42% (229/240) on Mobile (21.24% Auto CER), while both tiers invent 0 words across all 100 silent/noise clips.

Compressing the acoustic backbone by 7.0× into the 25.35 MiB Mobile Conformer-Small accelerates CPU inference to 0.0089× RTF (112× faster than real time; 0.0101× 1T RTF on held-out), while dynamic positional synthesis removes 9.77 MiB of static FP32 tables (35.61 → 25.85 MiB single-language first load; 31.78 MiB / 33.3 MB full 12-language + Auto-LID fleet, 10.7× smaller than Shruti's 356.2 MB). Table 2 details per-suite generalization across all 16 open benchmarks. On unseen Indic `FLEURS test` sets, Studio achieves 4.27%–8.83% CER across all ten Indian languages (4.27% Kannada, 4.56% Hindi, 5.13% Bengali, 5.51% Malayalam, 5.63% Gujarati, 6.76% Punjabi, 7.15% Odia, 7.88% Marathi, 8.00% Telugu, 8.83% Tamil), alongside 5.90% CER / 13.90% WER on `LibriSpeech test-clean` and 9.89% CER / 23.76% WER on `GLOBE` multi-accent English, while the 7× smaller Mobile student achieves 9.19% CER on Kannada, 13.56% on Hindi, 13.57% on Malayalam, 13.88% on Bengali, and 14.30% on `LibriSpeech test-clean`.

Table 3: Top: 12-Language 100 ms Streaming Telemetry ($N = 72$ sessions). Bottom: Multi-Accent, Slang & Reverberant-Mic Adaptation ($N = 150$, $B = 1{,}000$ bootstrap 95% CIs).

Streaming Metric ($N = 72$, $\Delta t = 100$ ms)	Mean	Median (p50)	p95 / Max	Target
Time-To-First-Token (TTFT)	30.51 ms	**28.62 ms**	50.81 ms (p95)	< 100 ms
Per-Chunk Decode Latency	**7.80 ms**	—	60,800 smp (buf)	≤ 61,440 smp
Streaming RTF / Committed Churn	0.0958×	—	**0 / 72 (0.00%)**	0.00% churn
Adaptation Slice ($N = 150$)	**Clips**	**Base CER / WER**	**Adapted CER [95% CI]**	**Adapted WER**
Global & South Asian Accents (`GLOBE`+`AMI`)	60	20.30% / 44.83%	**16.33% [12.06, 21.26]**	**30.28%**
Casual Contractions, Slang & Hinglish	36	28.07% / 74.80%	**20.91% [16.81, 26.02]**	**52.64%**
Indic Colloquial & English Loanwords	25	7.38% / 25.51%	**5.91% [3.40, 8.79]**	**17.85%**
Tempo & Reverberant Mobile-Mic Perturb.	29	20.42% / 50.54%	**24.41% [15.86, 33.85]**	**40.06%**
Overall Conversational Suite	**150**	20.04% / 49.90%	**17.26% [14.20, 20.54]**	**35.47%**

5.2 Streaming Stability and Multi-Accent / Slang Adaptation

Table 3 (top) reports streaming telemetry across $N = 72$ chunked 100 ms streaming sessions (1,600 samples per step) spanning all 12 languages. Bounding the rolling buffer at $W_{\max} = 61{,}440$ samples (3.84 s; maximum observed buffer 60,800 samples) and retaining bounded pre-rollover PCM for exact utterance-end CMVN parity holds mean per-step inference latency to 7.80 ms (streaming RTF 0.0958×), yielding 28.62 ms median Time-To-First-Token (mean 30.51 ms, p95 50.81 ms) and 0/72 (0.00%) committed-prefix churn violations across every script. On physical smartphone browsers (Android Chrome and iOS Safari), single-threaded WASM SIMD processes coalesced mobile windows in 57–149 ms (∼ 0.04× RTF) with zero dropped audio frames.

Table 3 (bottom) evaluates conversational robustness on $N = 150$ multi-accent, casual contraction/Hinglish, Indic loanword, and reverberant microphone clips. Combining causal Wiener + 80 Hz HPF + 50/60 Hz notch enhancement with pan-Indic + multi-accent Conformer adaptation lowers overall CER from 20.04% to 17.26% ([14.20%, 20.54%]) and WER from 49.90% to 35.47% ([31.19%, 39.58%]), cutting real human multi-accent WER to 30.28%, casual Hinglish/slang WER by 22.16 percentage points (74.80% → 52.64%), and Indic colloquial loanword CER to 5.91% (17.85% WER).

6 Discussion & Limitations

Spontaneous Meeting Overlap and Reverberant Recitation. Open-benchmark evaluation in Table 2 highlights two challenging acoustic regimes: spontaneous multi-party meetings (`ami_meeting_ihm`: 39.71% Studio / 55.49% Mobile CER) containing overlapping talkers, disfluencies, and sub-second backchannels, and crowd-recorded Classical Sanskrit (`sushrota_wild_test`: 18.80% Studio / 40.49% Mobile CER) featuring temple/hall reverberation and chanted *sandhi* compounds. While the 25.35 MiB Mobile

tier preserves strong intelligibility on clean and moderately noisy speech (`fleurs_kn` 9.19%, `fleurs_hi` 13.56%, `librispeech_clean` 14.30%), 4.9× fewer parameters limit robustness under severe reverberation and cross-talk.

Greedy Subword CTC vs. Agglutinative WER. Swanada uses zero-dependency greedy CTC decoding without an external n-gram or neural language model to keep per-language overhead at 515.0 KiB. In morphologically rich, agglutinative Dravidian scripts (`ml`, `ta`, `kn`, `te`) and Sanskrit *samāsa* compounds where a single whitespace-delimited word spans 12–25 characters, a single vowel-length or virama error flips the entire word, yielding 20.9%–37.3% Studio WER despite low 4.27%–8.83% CER. Optional compact KenLM fusion is a natural extension where memory budgets permit.

Sub-Millisecond Auto-LID Prefix Lock-In and Single-Threaded WebViews. Fusing the 395.8 KiB Attentive Statistics Pooling head (`lid.bin`) with top-8 CTC acoustic certainty eliminates cross-script character bleed and 12× softmax overhead while achieving 100.00% full-utterance Auto-LID on Studio and 95.42% on Mobile (< 1 ms overhead), though ultra-short < 2.4 s streaming prefixes with shared Indo-Aryan loanwords show lower early-prefix accuracy (85.00% Studio / 60.83% Mobile) before full-sentence confirmation. On iOS `WKWebView` browsers lacking `SharedArrayBuffer` support without Service Worker Jetsam risk, WASM runs single-threaded, making the 25.35 MiB Mobile tier essential for cool, responsive phone dictation.

7 Related Work

Multilingual and Indic Speech Recognition. Gulati et al. [7] introduced Conformers combining self-attention [15] and depthwise convolutions. Whisper [13] and Meta MMS [12] scaled multilingual ASR across dozens to over a thousand languages, while AI4Bharat Vistaar/IndicConformer [2] and Sushrota [1] established strong server-side baselines for Indian languages and Classical Sanskrit. Unlike monolithic PyTorch/NeMo checkpoints (180–600+ MB per model), CTranslate2 bilingual Whisper pairs lacking cross-Indic Auto-LID or working browser ONNX exports, or autoregressive decoders prone to looping on noisy browser audio, Swanada targets sub-30 MiB split-encoder WebAssembly execution with < 1 ms 12-way Auto-LID and hot-swappable 515 KiB native-script heads.

Knowledge Distillation and Browser Speech Systems. Hinton et al. [8] and Romero et al. [14] established logit and intermediate-representation distillation. Client-side speech toolkits such as `whisper.cpp`, `Transformers.js`, and `sherpa-onnx` bring quantized models to WebAssembly, while Ninada [9] introduced runtime binary ONNX graph splitting and capability-routed mobile/studio specialists for 24 kHz Indic TTS. Swanada extends this split-graph, capability-routed paradigm to 16 kHz streaming ASR via zero-table relative-PE synthesis, `MatMul`-backed `QInt8` Conformer blocks, post-quantization head calibration, single-pass Attentive Statistics Pooling Auto-LID, and zero-churn LCP stabilization.

8 Conclusion

We presented **Swanada**, an open-source client-side speech recognition system spanning Classical Sanskrit, English, and ten Indian languages across browser WebAssembly, Node.js, and Python. By combining < 2 ms binary ONNX `GraphProto` tail surgery, twelve hot-swappable 515.0 KiB native-script CTC heads, a 395.8 KiB 12-way Auto-LID head (100.0% Studio / 95.4% Mobile accuracy), a distilled 25.35 MiB Mobile Conformer-Small alongside a 177.38 MiB Studio Conformer-Large, and two-zone LCP streaming stabilization, Swanada achieves 5.35% held-out CER, 8.25% 12-language open `FLEURS`+Wild CER (19.23% Mobile) and 10.44% 16-suite CER across 1,600 public `test` clips, 24.18% 10-Indic `FLEURS` WER (10/10 wins vs. Shruti's 32.20%), 0/100 silent hallucinations, 0.009×–0.021× CPU RTF, and 28.6 ms median streaming TTFT with 0.00% committed-prefix churn. Source code, reproducible benchmark harnesses, and open weights are available at https://github.com/hemanth/swanada.

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